



EmotiCare

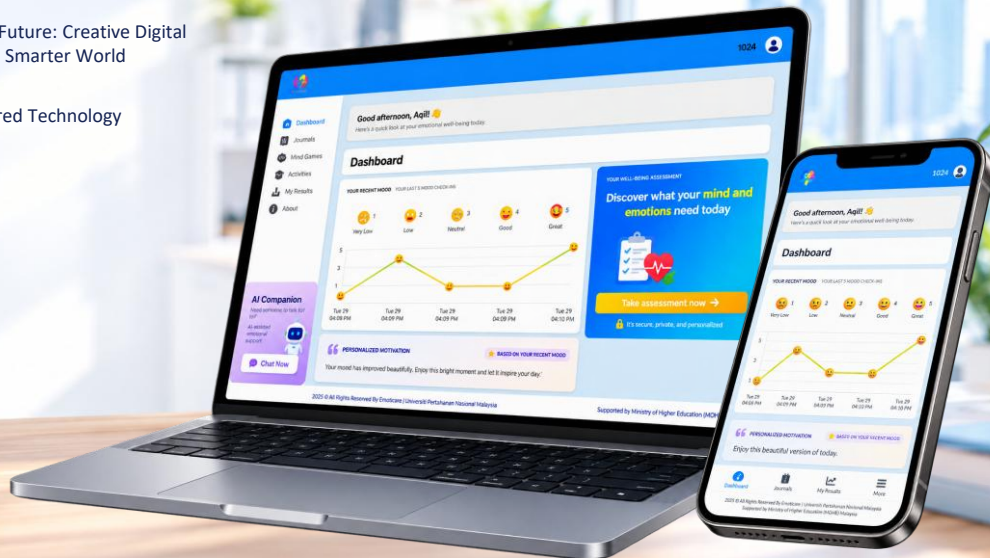
An Intelligence Mental Health Conversational Assistance Platform

Theme

Designing the Future: Creative Digital Solutions for a Smarter World

Subtheme

Human-Centered Technology



Project Overview

EmotiCare is a web-based emotional and mental health support platform designed to provide accessible, user-centered digital support. The platform integrates emotional assessments, mood check-ins, journaling, emotional analysis, wellness activities and personalized support within a single web environment.

The platform is designed to place users at the center of the digital experience by combining accessible web technology with artificial intelligence and emotional intelligence approaches to support emotional self-awareness and early support.



Mood Check-In



Emotional Assessments



Journaling



Wellness Activities



Personalized Insights



AI Conversational Assistant

The Problem



Fragmented Digital Support

Mental and emotional support activities can be distributed across different tools and services, making the user experience less centralized.



Limited Emotional Self-Monitoring

Users may need a simple way to regularly reflect on their mood, emotional state and personal experiences.



Need for Personalized Digital Support

Different users may experience different emotional needs, creating a need for technology that can provide more context-aware and personalized support.

Target Users



Students

Support emotional self-awareness, assessment and wellness activities.



Staff

Provide accessible tools for emotional check-ins and personal reflection.



Organisations

Provide a digital platform that can support employee/student emotional well-being initiatives.



EmotiCare brings emotional assessment, monitoring, AI-assisted analysis and wellness support together in one human-centered web platform.

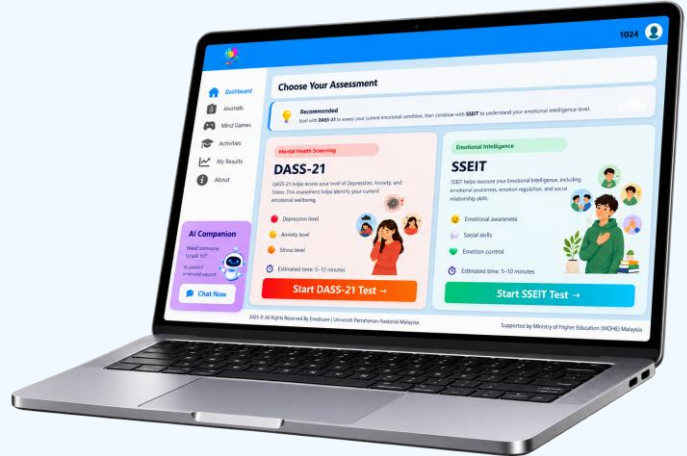
Our Solution & Key Features

EmotiCare provides an integrated web-based platform that combines emotional assessment, mood monitoring, journaling, emotional analysis and wellness activities in one accessible environment. The platform is designed to help users understand their emotional state, reflect on their experiences and access appropriate digital support.

01 Emotional Assessments

Provides multiple psychological and emotional assessment instruments to help users understand different aspects of their emotional and psychological profile.

- ✓ DASS-21
- ✓ SSEIT



02 Mood Check-In

Users can record their current emotional state through a simple and engaging mood check-in interface.

- ✓ Simple and quick
- ✓ Track mood patterns
- ✓ Build emotional self-awareness



03 Digital Journaling

Users can record their thoughts and personal experiences through the journaling module. The platform can perform AI-assisted emotional analysis on journal content.

- ✓ Encourage self-reflection
- ✓ Identify emotional patterns
- ✓ Support early identification of negative emotional signals



04 AI-Assisted Emotional Analysis

EmotiCare integrates AI/ML techniques to analyse textual and emotional content.

- ✓ Language detection (EN, BM, etc.)
- ✓ VADER sentiment analysis
- ✓ TF-IDF feature extraction
- ✓ Logistic regression classification
- ✓ Emotion detection
- ✓ Intervention type

05 Wellness Activities

Users can access digital activities designed to encourage emotional engagement and well-being.

- ✓ Mood Match
- ✓ Memory Focus
- ✓ Fitness Activities
- ✓ Music Therapy
- ✓ Reading



06 Support & Intervention

- ✓ Email notifications for defined alert conditions

- ✓ When the system identifies situations requiring additional attention, EmotiCare can provide appropriate support pathways.

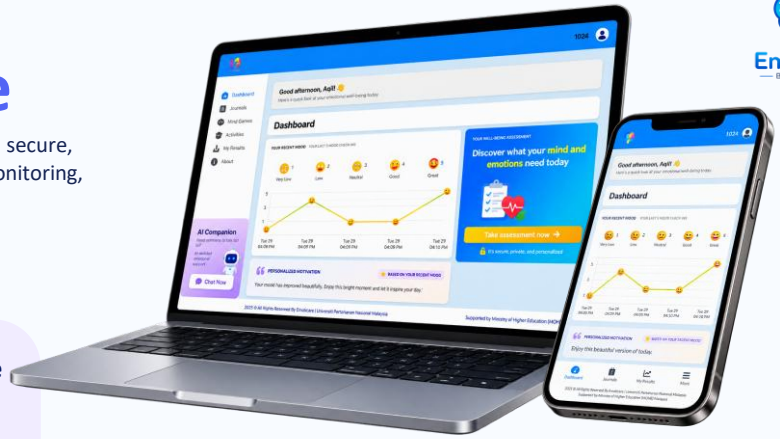
How EmotiCare Works

EmotiCare connects multiple emotional-care activities into a continuous digital experience, allowing users to move from self-assessment and emotional reflection to insights and appropriate support.



Technology & System Architecture

EmotiCare combines web technologies with AI/ML to provide a secure, scalable and integrated platform for emotional assessment, monitoring, analysis and digital support.



Web Application

PHP + HTML + CSS + JavaScript

- ✓ User Interface
- ✓ Navigation
- ✓ Assessment modules
- ✓ Journaling
- ✓ Mood tracking
- ✓ Result pages
- ✓ Activity modules

Database

MySQL

- ✓ Users
- ✓ Assessment results
- ✓ Mood records
- ✓ Journals
- ✓ Activities
- ✓ Counsellors
- ✓ System activity

AI / Machine Learning

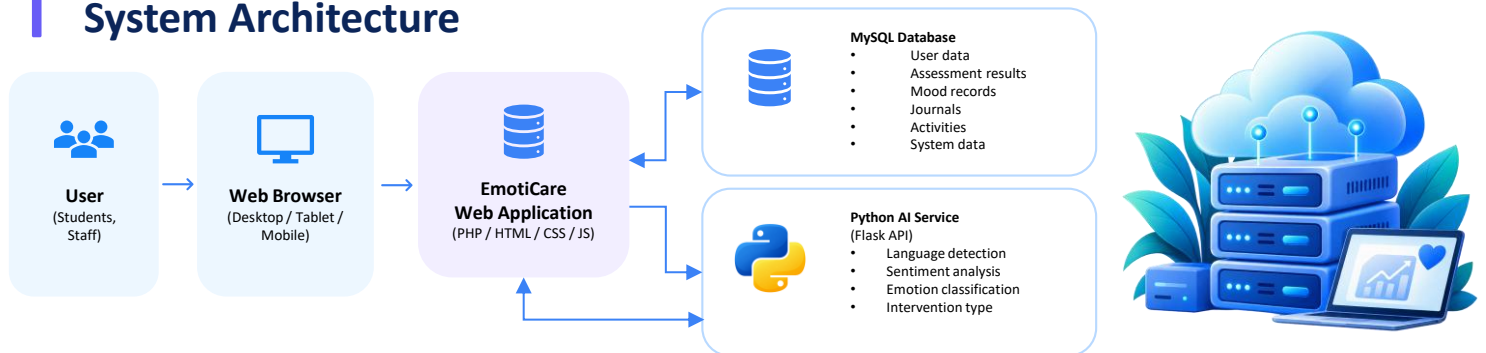
Python + Flask

- ✓ Language detection (Lingua)
- ✓ VADER sentiment analysis
- ✓ TF-IDF feature extraction
- ✓ Logistic regression classification
- ✓ Emotion detection
- ✓ Intervention type

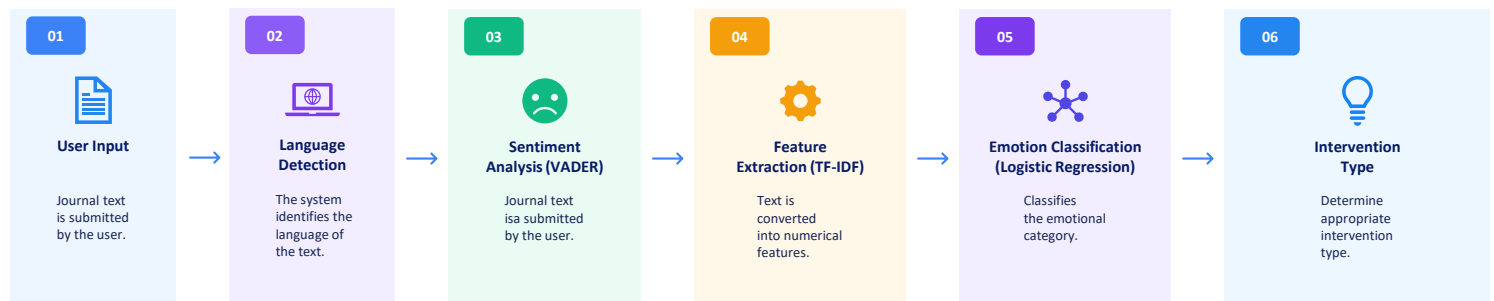
Security & Authentication

- ✓ PHP session management
- ✓ Passkey / WebAuths
- ✓ Password protection
- ✓ Account verification
- ✓ Login attempts controls
- ✓ Session timeout (15 minutes)
- ✓ Encrypted journal content

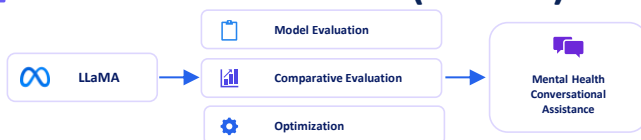
System Architecture



AI Emotional Analysis Pipeline



Research Direction (LLaMA)



Automated Evaluation

- BLEU
- ROUGE-N
- BERTScore

Human-Centred Evaluation

- Empathy
- Appropriateness
- Consistency

EmotiCare brings emotional assessment, monitoring, AI-assisted analysis and wellness support together in one human-centered web platform.

UI/UX & Human-Centered Design

EmotiCare uses a consistent and user-centered interface to make emotional assessment, monitoring and wellness activities easier to access and understand. The platform is designed around the user's emotional journey with clear navigation, consistent components and a responsive experience.



Clear Navigation

Users can move between major modules through a consistent navigation structure.



Consistent Components

Frequently used interface elements follow a consistent visual and interaction pattern across the platform.



Responsive Experience

The interface adapts to different screen sizes for desktop, tablet and mobile users.



User-Friendly Interaction

Information is presented through clear cards, forms, buttons, icons and visual feedback to reduce unnecessary complexity.

Emotional Accessibility

EmotiCare presents emotional-care functions in a simple and approachable way to reduce barriers when users interact with the platform.



Simple Language

Clear and understandable wording supports easier interaction.



Visual Guidance

Icons, colours and visual cues help users recognize functions quickly.



Low-Complexity Interaction

Simple actions reduce unnecessary steps when accessing emotional-care features.



Supportive Feedback

Friendly interface feedback creates a more supportive digital experience.

Responsive Design

The interface is optimised for desktop, tablet and mobile devices, allowing the same core functionality to remain accessible without requiring a separate mobile application.



Desktop

Tablet

Mobile

Human-Centered Emotional Experience

EmotiCare is designed around the user's emotional journey rather than technology alone.



Check Mood
Record current emotional state



Assessment
Complete emotional assessments



Reflect & Journal
Write and reflect on experiences



Understand Results
View insights and progress



Access Support
Wellness activities and counselling support

Key Interface Screenshots

Consistent interaction patterns, responsive layouts and clear information presentation support a coherent user experience across the platform.

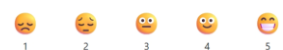
01 Dashboard



Overview of key modules, emotional well-being insights and quick access to main features.

02 Mood Check-In

How do you feel now?



A simple and engaging interface for users to record their current emotional state.

03 Result



Clear visualization of assessment results to help users understand their emotional state.

04 Journal



A clean and user-friendly interface for journaling.

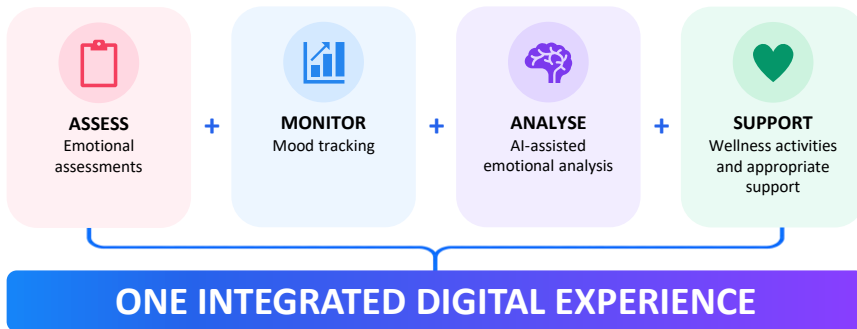


Thoughtful design. Meaningful experience. A more supportive tomorrow with EmotiCare.

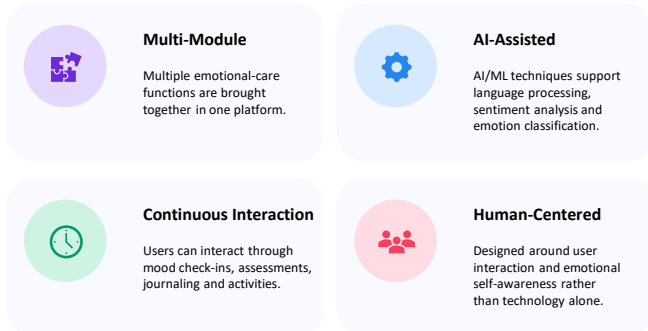
Innovation, Impact & Future Development

EmotiCare integrates emotional assessment, mood monitoring, journaling, AI-assisted emotional analysis and wellness activities within a single-web-based platform to support emotional well-being in a more accessible and connected digital experience.

Integrated Human-Centered Digital Care



What Makes EmotiCare Different



Potential Impact

EmotiCare can support emotional well-being initiatives across different communities.



Development Roadmap

A phased approach to enhance capabilities and expand impact.



EmotiCare aims to combine intelligent digital technology with human-centred emotional support to create a more accessible and connected digital care experience.



AI Usage Disclosure

Generative AI was used as a supporting tool during the development and preparation process. The system concept, implementation, integration, testing and final design decisions were developed and validated by the project team.